

PRESS SUMMARY

Security in Decentralized Blockchain-Based Messaging Systems

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THE PROBLEM

Today's most widely used messaging platforms — WhatsApp, Messenger, Telegram — rely on centralized servers operated by large corporations. When those servers go down, millions of people simultaneously lose access to their communications. In 2019, a Facebook outage left 2.3 billion users without service for more than 14 hours. Beyond reliability, centralized systems are also vulnerable to malicious attacks, government surveillance, and data interception.

THE PROPOSAL

This paper presents an alternative messaging system built on blockchain technology — the same technology that powers cryptocurrencies like Bitcoin. Instead of routing messages through a central server, messages travel directly between users across a distributed network. Each message is encrypted using public-key cryptography before it is sent, ensuring that only the intended recipient can read it, even if the content is visible on the network. No company or centralized server needs to be involved in the communication.

HOW IT WORKS

Each message is encrypted with the recipient's public key and stored inside a block on the chain. Every block is validated using SHA-256, the same security algorithm used in international banking transactions. The identities of sender and recipient remain anonymous within the network. The system runs in a standard web browser with no additional software installation required.

IDENTIFIED LIMITATIONS

The research also documents the current challenges of this technology: transaction verification speed is slower than in centralized systems, which limits short-term mass adoption. Known vulnerabilities were also identified — the 51% attack, where a single actor accumulates enough computing power to control the network, and the double-spending problem, both inherent to public blockchains. The paper proposes specific solutions to each of these challenges.

SIGNIFICANCE

The research was selected for publication in IEEE Xplore, the world's leading scientific database in electrical engineering and technology, which publishes only peer-reviewed work. It was presented at CONIITI 2021, an international congress organized by Universidad Católica de Colombia with researchers from more than 15 countries.

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FULL PAPER

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